



Non-Equilibrium Thermodynamics: The Multispecies Boltzmann Equation

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Multispecies Boltzmann Equation

A Fundamental Identity



In the last slide deck, we proved the following:

$$\begin{aligned}
 & \int_{\mathbb{R}^3} Q_{ij}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) \, d\mathbf{v} \\
 &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi - \psi'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\
 &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi_* - \psi'_*] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}).
 \end{aligned}$$

This is, perhaps, the most fundamental result for the Boltzmann equation, as we shall see.



Proposition

Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ is given and integrable. Then, assuming the collision kernel $B_{i,i}$ has proper structure, it follows that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \, d\mathbf{v} = 0, \quad (1)$$

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} = \mathbf{0}, \quad (2)$$

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) |\mathbf{v}|^2 \, d\mathbf{v} = 0. \quad (3)$$



Proof.

These results follow from identity

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi + \psi_* - \psi' - \psi'_*] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \end{aligned} \quad (4)$$

and the transformation invariants (binary collision invariants)

$$m_i \mathbf{v} + m_j \mathbf{v}_* = m_i \mathbf{v}' + m_j \mathbf{v}'_* \quad (5)$$

$$\frac{m_i}{2} |\mathbf{v}|^2 + \frac{m_j}{2} |\mathbf{v}_*|^2 = \frac{m_i}{2} |\mathbf{v}'|^2 + \frac{m_j}{2} |\mathbf{v}'_*|^2. \quad (6)$$

for the case that $i = j$. ■



Definition

Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ is given and integrable. Any function $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ with the property that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) \, d\mathbf{v} = 0 \quad (7)$$

is called a **species i collisional invariant** or a **unary collisional invariant**.

Clearly, if the collision kernel $B_{i,i}$ has proper structure, then $\psi \equiv 1$, $\psi(\mathbf{v}) = \mathbf{v}$, and $\psi(\mathbf{v}) = |\mathbf{v}|^2$ are unary collisional invariants.



Proposition

For $i \neq j$, if $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are given and integrable, then, assuming the collision kernel $B_{i,j}$ has proper structure, it follows that

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \, d\mathbf{v} = 0, \quad (8)$$

$$m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} = \mathbf{0}, \quad (9)$$

$$m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) |\mathbf{v}|^2 \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) |\mathbf{v}|^2 \, d\mathbf{v} = 0. \quad (10)$$



Proof.

The identity (8) follows from

$$\begin{aligned}
 & \int_{\mathbb{R}^3} Q_{ij}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v} \\
 &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi - \psi'] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\
 &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi_* - \psi'_*] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \\
 & \hspace{15em} (11)
 \end{aligned}$$

with $\psi \equiv 1$.



Proof. (Cont.)

Using $\psi(\mathbf{v}) = \mathbf{v}$ in (11), we find that

$$\begin{aligned}
 & m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \\
 &= \frac{m_i}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v} - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\
 &+ \frac{m_i}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v}_* - \mathbf{v}'_*] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \\
 &+ \frac{m_j}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v} - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \\
 &+ \frac{m_j}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v}_* - \mathbf{v}'_*] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}) \\
 &= \mathbf{0}. \tag{12}
 \end{aligned}$$



Proof. (Cont.)

We have used the following binary momentum invariants:

$$m_i \mathbf{v} + m_j \mathbf{v}_* = m_i \mathbf{v}' + m_j \mathbf{v}'_* \iff m_i (\mathbf{v} - \mathbf{v}') = m_j (\mathbf{v}'_* - \mathbf{v}_*) \quad (' \leftrightarrow T_{\sigma}^{i,j}), \quad (13)$$

$$m_j \mathbf{v} + m_i \mathbf{v}_* = m_j \mathbf{v}' + m_i \mathbf{v}'_* \iff m_j (\mathbf{v} - \mathbf{v}') = m_i (\mathbf{v}'_* - \mathbf{v}_*) \quad (' \leftrightarrow T_{\sigma}^{j,i}). \quad (14)$$

Using $\psi(\mathbf{v}) = |\mathbf{v}|^2$ in (11) and the appropriate binary kinetic energy invariants, the identity (10) immediately follows. ■



Proposition

Suppose that $i, j \in \{1, \dots, R\}$ are arbitrary and $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are given and integrable. Assume that the collision kernel B_{ij} has proper structure. Recall our definition

$$Q_i[f_i, \mathbf{f}](\mathbf{v}) := \sum_{j=1}^R Q_{ij}[f_i, f_j](\mathbf{v}). \quad (15)$$

Then,

$$\int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) \, d\mathbf{v} = 0, \quad (16)$$

$$\sum_{i=1}^R m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) \mathbf{v} \, d\mathbf{v} = \mathbf{0}, \quad (17)$$

$$\sum_{i=1}^R m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) |\mathbf{v}|^2 \, d\mathbf{v} = 0. \quad (18)$$



Proof.

We prove only (17). The other two identities are proved similarly. Observe that

$$\begin{aligned}
 J &= \sum_{i=1}^R m_i \int_{\mathbb{R}^3} \sum_{j=1}^R Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \\
 &= \sum_{i=1}^R \sum_{j=1}^R m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \\
 &= \sum_{i=2}^R \sum_{j=1}^i \left(m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \right) \\
 &\quad + \sum_{i=1}^R m_i \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \\
 &= \mathbf{0}.
 \end{aligned}$$





With the next result, we show that all unary collisional invariants have special structure.

Proposition

Suppose that $\psi \in C^2(\mathbb{R}^3; \mathbb{R})$ has the property that

$$\psi(\mathbf{v}) + \psi(\mathbf{v}_*) = \psi(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) + \psi(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)), \quad (19)$$

for all $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$ and all $\sigma \in \mathbb{S}^2$. Then, there exist constants $A, C \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, such that

$$\psi(\mathbf{v}) = \frac{m_i A}{2} |\mathbf{v}|^2 + m_i \mathbf{B} \cdot \mathbf{v} + C. \quad (20)$$

In other words, ψ is a velocity-space quadratic function.

Proof.

Define

$$\frac{m_i A}{2} = \frac{1}{6} \Delta \psi(\mathbf{0}),$$

$$m_i \mathbf{B} = \nabla \psi(\mathbf{0}),$$

$$C = \psi(\mathbf{0}).$$



Proof (Cont.)

Set

$$\phi(\mathbf{v}) = \psi(\mathbf{v}) - \left(\frac{m_i A}{2} |\mathbf{v}|^2 + m_i \mathbf{B} \cdot \mathbf{v} + C \right), \quad \forall \mathbf{v} \in \mathbb{R}^3.$$

Then $\phi \in C^2(\mathbb{R}^3; \mathbb{R})$ is a unary collisional invariant, that is,

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \phi(\mathbf{v}) \, d\mathbf{v} = 0,$$

with the further property that

$$\phi(\mathbf{0}) = 0, \quad \nabla \phi(\mathbf{0}) = \mathbf{0}, \quad \Delta \phi(\mathbf{0}) = 0.$$

Now, suppose that $\mathbf{v}_* \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ is fixed but, otherwise, arbitrary. Suppose $\boldsymbol{\sigma} \in \mathbb{S}^2$ is perpendicular to $\mathbf{v}_*$, that is,

$$\boldsymbol{\sigma} \cdot \mathbf{v}_* = 0.$$



Proof (Cont.)

Set, for $s \in \mathbb{R} \setminus \{0\}$,

$$\mathbf{v} = s\boldsymbol{\sigma}.$$

Then, under the transformation $T_{\boldsymbol{\sigma}}^{i,i}$,

$$\begin{aligned}\mathbf{v}' &:= T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}, \mathbf{v}_*) \\ &= \mathbf{v} - \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*) \frac{2m_i}{m_i + m_i} \boldsymbol{\sigma} \\ &= \mathbf{v} - s\boldsymbol{\sigma} \\ &= \mathbf{0}.\end{aligned}$$

Furthermore,

$$\mathbf{v}'_* := T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}, \mathbf{v}_*) = \mathbf{v}_* + s\boldsymbol{\sigma}.$$

Proof (Cont.)



Thus,

$$\begin{aligned}
 \phi(s\sigma) + \phi(\mathbf{v}_*) &= \phi(\mathbf{v}) + \phi(\mathbf{v}_*) \\
 &= \phi(\mathbf{v}') + \phi(\mathbf{v}'_*) \\
 &= \phi(\mathbf{0}) + \phi(\mathbf{v}_* + s\sigma) \\
 &= \phi(\mathbf{v}_* + s\sigma).
 \end{aligned}$$

Equivalently,

$$\phi(s\sigma) = \phi(\mathbf{v}_* + s\sigma) - \phi(\mathbf{v}_*).$$

Using l'Hôpital's rule and the fact that $\phi(\mathbf{0}) = 0$ and $\nabla\phi(\mathbf{0}) = \mathbf{0}$, it follows that

$$\sigma \cdot \nabla\phi(\mathbf{v}_*) = 0.$$

Since $\mathbf{v}_* \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ and $\sigma \in \mathbb{S}^2$ are perpendicular but otherwise arbitrary, it must be that

$$\phi(\mathbf{v}) = g(|\mathbf{v}|), \quad \forall \mathbf{v} \in \mathbb{R}^3,$$

that is, ϕ is radially symmetric.



Proof (Cont.)

Next, assume that $\mathbf{n} \in \mathbb{S}^2$ is arbitrary. Set

$$\mathbf{v} = s\mathbf{n} \quad \text{and} \quad \mathbf{v}_* = r\mathbf{n},$$

where

$$0 \leq s < r.$$

Pick $\boldsymbol{\sigma} \in \mathbb{S}^3$ such that

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \pm \frac{1}{\sqrt{2}}.$$



Proof (Cont.)

With this particular choice,

$$\mathbf{v}' = s\mathbf{n} - (s - r)\frac{\pm 1}{\sqrt{2}}\boldsymbol{\sigma},$$

$$\mathbf{v}'_{\star} = r\mathbf{n} + (s - r)\frac{\pm 1}{\sqrt{2}}\boldsymbol{\sigma},$$

and

$$|\mathbf{v}'|^2 = \frac{s^2 + r^2}{2},$$

$$|\mathbf{v}'_{\star}|^2 = \frac{s^2 + r^2}{2}.$$



Proof (Cont.)

Since

$$\phi(\mathbf{v}) + \phi(\mathbf{v}_*) = \phi(\mathbf{v}') + \phi(\mathbf{v}'_*),$$

and ϕ is radially symmetric, we have

$$g(s) + g(r) = 2g\left(\sqrt{\frac{s^2 + r^2}{2}}\right).$$

It follows that, upon differentiating with respect to s ,

$$\begin{aligned} g''(s) &= g''\left(\sqrt{\frac{s^2 + r^2}{2}}\right) \sqrt{\frac{2}{s^2 + r^2}} s + g'\left(\sqrt{\frac{s^2 + r^2}{2}}\right) \sqrt{\frac{2}{s^2 + r^2}} \\ &\quad - g'\left(\sqrt{\frac{s^2 + r^2}{2}}\right) \left(\frac{2}{s^2 + r^2}\right)^{3/2} \frac{s^2}{2}. \end{aligned}$$



Proof (Cont.)

Finally, for any $r > 0$,

$$0 = g''(s)|_{s=0} = g' \left(\frac{r}{\sqrt{2}} \right) \frac{\sqrt{2}}{r},$$

which implies that $g' \left(\frac{r}{\sqrt{2}} \right) = 0$, for any $r > 0$. The only possibility is that $g \equiv 0$.



Entropy, and Boltzmann's H-Theorem



Definition

Suppose that $f_1, \dots, f_R : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the multi-species Boltzmann equation. The quantity $s : \Omega \times [0, T] \rightarrow \mathbb{R}$ defined via

$$s(\mathbf{x}, t) := - \sum_{i=1}^R \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v},$$

is called the **entropy density** of the multi-species mixture. The quantity

$$S(t) = \int_{\Omega} s(\mathbf{x}, t) \, d\mathbf{x}$$

is called the **total entropy** of the mixture.



We will need the following technical lemma.

Lemma

For any $x, y \in (0, \infty)$,

$$(\ln(x) - \ln(y))(x - y) \geq 0, \quad (21)$$

with equality iff $x = y$.

Proof.

Observe that the function $q(x) = x \ln(x)$ is strictly convex on $[0, \infty)$. In fact, for any $x \in (0, \infty)$,

$$q''(x) = \frac{1}{x} > 0.$$

By Taylor's theorem, for any $x, y \in (0, \infty)$, $x \neq y$,

$$x \ln(x) = y \ln(y) + (\ln(y) + 1)(x - y) + \frac{1}{2} \frac{1}{\xi} (x - y)^2 > y \ln(y) + (\ln(y) + 1)(x - y),$$

for some ξ between x and y .

Proof (Cont.)



The last inequality can be rewritten as

$$x \ln(x) - x \ln(y) > x - y.$$

Reversing the roles of x and y , we have

$$y \ln(y) - y \ln(x) > y - x.$$

Adding the inequalities, we have

$$(x - y)(\ln(x) - \ln(y)) > 0. \quad (22)$$

If $x = y$, then, clearly,

$$(\ln(x) - \ln(y))(x - y) = 0.$$

Conversely, suppose that

$$(\ln(x) - \ln(y))(x - y) = 0.$$

We get a clear contradiction, unless $x = y$, because of the strict inequality (22). ■



Proposition (H-Theorem)

Suppose that $f_1, \dots, f_R : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the multi-species Boltzmann equation with specular reflection-type boundary conditions. Assume that the collision kernels $B_{i,j}$ have proper structure. Then,

$$\partial_t s(\mathbf{x}, t) + \nabla_{\mathbf{x}} \cdot \mathbf{J}_s(\mathbf{x}, t) = \sigma_s(\mathbf{x}, t) \geq 0, \quad (23)$$

where $\mathbf{J}_s$ is the **entropy flux**, given by

$$\mathbf{J}_s := - \sum_{i=1}^R \int_{\mathbb{R}^3} \mathbf{v} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v},$$

and σ_s is the **entropy production rate**, defined via

$$\sigma_s(\mathbf{x}, t) := - \sum_{i=1}^R \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v}.$$



Proposition (H-Theorem (Cont.))

The total entropy satisfies

$$d_t S(t) = \int_{\Omega} \sigma_s(\mathbf{x}, t) d\mathbf{x} \geq 0, \quad (24)$$

since

$$\mathbf{n}(\mathbf{x}) \cdot \mathbf{J}_s(\mathbf{x}, t) = 0, \quad \forall \mathbf{x} \in \partial\Omega. \quad (25)$$

Proof.

Let us multiply the i^{th} equation of multi-species Boltzmann system by $\ln(f_i) + 1$, integrate in velocity space, and sum on i . We clearly obtain (23) after using

$$\int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) d\mathbf{v} = 0.$$



Proof (Cont.)

Next, we want to show that

$$\sigma_s(\mathbf{x}, t) = - \sum_{i=1}^R \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} \geq 0, \quad \forall \mathbf{x} \in \Omega, \quad \forall t \geq 0.$$

Let us write this as the sum of sums, a diagonal sum and an off-diagonal sum:

$$\begin{aligned} -\sigma_s &= \sum_{i=1}^R \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &\quad + \sum_{i=2}^R \sum_{j=1}^i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \sum_{j=2}^R \sum_{i=1}^j \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} \\ &= \sum_{i=1}^R \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &\quad + \sum_{i=2}^R \sum_{j=1}^i \left[\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \right]. \quad (26) \end{aligned}$$

Proof (Cont.)



Let us tackle the diagonal terms first. Observe that, using (4),

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i + L_{i,*} - L'_i - L'_{i,*}] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ & \quad (i \leftrightarrow j), \end{aligned} \quad (27)$$

where

$$L_i := \ln(f_i), \quad L_{i,*} := \ln(f_{i,*}), \quad L'_i = \ln(f'_i), \quad L'_{i,*} = \ln(f'_{i,*}).$$

Now, observe that

$$L_i + L_{i,\star} - L'_i - L'_{i,\star} = \ln(f_i f_{i,\star}) - \ln(f'_i f'_{i,\star}).$$

Using estimate (21), it follows that

$$[f'_i f'_{i,*} - f_i f_{i,*}] [L_i + L_{i,*} - L'_i - L'_{i,*}] \leq 0,$$

for each $\mathbf{x} \in \Omega$, every $\mathbf{v} \in \mathbb{R}^3$, and every $t \geq 0$.



Proof (Cont.)

Since $B_{i,i}(\mathbf{w}, \boldsymbol{\sigma}) \geq 0$, for all $\mathbf{w} \in \mathbb{R}^3$ and $\boldsymbol{\sigma} \in \mathbb{S}^2$, it follows that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) d\mathbf{v} \leq 0,$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. As a consequence,

$$\sum_{i=1}^R \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) d\mathbf{v} \leq 0.$$



Proof (Cont.)

Next we address the off-diagonal sum in (26). Using (11), for $i > j$,

$$\begin{aligned}
 & \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) \, d\mathbf{v} \\
 &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i - L'_i] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}) \\
 &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{i,*} - L'_{i,*}] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}),
 \end{aligned}$$

and

$$\begin{aligned}
 & \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) \, d\mathbf{v} \\
 &= \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_j - L'_j] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}) \\
 &+ \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{j,*} - L'_{j,*}] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}).
 \end{aligned}$$



Proof (Cont.)

Thus, for $i > j$,

$$\begin{aligned}
 & \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) \, d\mathbf{v} \\
 &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i - L'_i + L_{j,*} - L'_{j,*}] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{i,*} - L'_{i,*} + L_j - L'_j] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}
 \end{aligned} \tag{28}$$

Using the logarithmic estimate (21) and the fact that $B_{i,j}, B_{j,i} \geq 0$, we conclude that

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) \, d\mathbf{v} \leq 0.$$

Both terms in (26) are non-positive, and we conclude that $\sigma_h \geq 0$. The estimate (23) is proven.

To prove (24), integrate (23) over Ω , use the Divergence Theorem, the specular boundary condition, and the fact that $\sigma_s \geq 0$, point-wise. ■



Proposition

Suppose that $f_1, \dots, f_R : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions. Assume that the collision kernels $B_{i,j}$ have proper structure. Then,

$$-\sigma_s(\mathbf{x}, t) = \sum_{i=1}^R \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} = 0 \quad (29)$$

iff, for $i = 1, \dots, R$,

$$\ln(f_i(\mathbf{x}, \mathbf{v}, t)) = \frac{m_i A(\mathbf{x}, t)}{2} |\mathbf{v}|^2 + m_i \mathbf{B}(\mathbf{x}, t) \cdot \mathbf{v} + C_i(\mathbf{x}, t), \quad (30)$$

where A and $\mathbf{B}$ are independent of i and $\mathbf{v}$.



Proof.

We have seen from the proof of the last proposition that

$$\begin{aligned}
 -\sigma_s &= \sum_{i=1}^R \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\
 &\quad + \sum_{i=2}^R \sum_{j=1}^i \left[\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \right] \\
 &\leq 0.
 \end{aligned} \tag{31}$$

More specifically,

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \leq 0, \quad i \in \{1, \dots, R\},$$

and

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \leq 0, \quad i, j \in \{1, \dots, R\}, \quad i > j.$$



Non-Equilibrium Thermodynamics



Proof (Cont.)

Conversely, if $\sigma_h(\mathbf{x}, t) = 0$ at a particular point $(\mathbf{x}, t) \in \Omega \times [0, \infty)$, then it must be that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} = 0, \quad \forall i \in \{1, \dots, R\}, \quad (32)$$

and

$$\int_{\mathbb{R}^3} Q_{ij}[f_i, f_j] \ln(f_i) d\mathbf{v} + \int_{\mathbb{R}^3} Q_{ji}[f_j, f_i] \ln(f_j) d\mathbf{v} = 0, \quad \forall i, j \in \{1, \dots, R\}, \quad i > j. \quad (33)$$



Proof (Cont.)

Using Equation (32), Equation (27), and a previous lemma, it must be that

$$L_j + L_{j,*} - L'_j - L'_{j,*} = 0 \quad (j \leftrightarrow T_{\sigma}^{i,i}),$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and $\boldsymbol{\sigma}$. Invoking a previous proposition, it follows that, for $i = 1, \dots, R$, for all $\mathbf{v} \in \mathbb{R}^3$,

$$L_i = \ln(f_i(\mathbf{x}, \mathbf{v}, t)) = \frac{m_i A_i(\mathbf{x}, t)}{2} |\mathbf{v}|^2 + m_i \mathbf{B}_i(\mathbf{x}, t) \cdot \mathbf{v} + C_i(\mathbf{x}, t). \quad (34)$$

Here, A_i and $\mathbf{B}_i$ are independent of $\mathbf{v}$, but we leave open the possibility that A_i and $\mathbf{B}_i$ depend upon i .



Proof (Cont.)

Next, using Equation (33), Equation (28), and a previous lemma, it must be that

$$L_i - L'_i + L_{j,*} - L'_{j,*} = 0 \quad (' \leftrightarrow T_{\sigma}^{i,j})$$

and

$$L_{i,*} - L'_{i,*} + L_j - L'_j = 0 \quad (' \leftrightarrow T_{\sigma}^{j,i}),$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ . We now insert (34)—dropping the $\mathbf{x}$ and t dependencies, since these points are fixed—into the last two equations, to obtain

$$\begin{aligned} & \frac{m_i A_i}{2} |\mathbf{v}|^2 - \frac{m_i A_i}{2} |\mathbf{v}'|^2 + \frac{m_j A_j}{2} |\mathbf{v}_*|^2 - \frac{m_j A_j}{2} |\mathbf{v}'_*|^2 \\ & + m_i \mathbf{B}_i \cdot \mathbf{v} - m_i \mathbf{B}_i \cdot \mathbf{v}' + m_j \mathbf{B}_j \cdot \mathbf{v}_* - m_j \mathbf{B}_j \cdot \mathbf{v}'_* = 0 \quad (' \leftrightarrow T_{\sigma}^{i,j}) \end{aligned}$$

and

$$\begin{aligned} & \frac{m_i A_i}{2} |\mathbf{v}_*|^2 - \frac{m_i A_i}{2} |\mathbf{v}'_*|^2 + \frac{m_j A_j}{2} |\mathbf{v}|^2 - \frac{m_j A_j}{2} |\mathbf{v}'|^2 \\ & + m_i \mathbf{B}_i \cdot \mathbf{v}_* - m_i \mathbf{B}_i \cdot \mathbf{v}'_* + m_j \mathbf{B}_j \cdot \mathbf{v} - m_j \mathbf{B}_j \cdot \mathbf{v}' = 0 \quad (' \leftrightarrow T_{\sigma}^{j,i}), \end{aligned}$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ .



Proof (Cont.)

Now, we invoke the particle transformation invariances, (5) and (6), to obtain

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}_\star|^2 - \frac{m_j}{2} |\mathbf{v}'_\star|^2 \right) + (\mathbf{B}_j - \mathbf{B}_i) \cdot (m_j \mathbf{v}_\star - m_j \mathbf{v}'_\star) = 0 \quad ({}' \leftrightarrow T_\sigma^{ij}), \quad (35)$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ , and

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}|^2 - \frac{m_j}{2} |\mathbf{v}'|^2 \right) + (\mathbf{B}_j - \mathbf{B}_i) \cdot (m_j \mathbf{v} - m_j \mathbf{v}') = 0 \quad (i \leftrightarrow j), \quad (36)$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ .



Proof (Cont.)

Let us examine Equation (35). Recall, from the transformation T_{σ}^{ij} ,

$$\mathbf{v}'_{\star} = \mathbf{v}_{\star} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_{\star})] \mu_{ij} \boldsymbol{\sigma}, \quad \mu_{ij} := \frac{2m_i}{m_i + m_j}.$$

Let $\mathbf{v}_\star$ be an arbitrary, non-zero vector. Choose

$$\mathbf{v} = \frac{\mu_{i,j} - 2}{\mu_{i,j}} \mathbf{v}_*, \quad \sigma = \frac{\mathbf{v}_*}{\|\mathbf{v}_*\|}.$$

Then, it follows that $\mathbf{v}'_{\star} = -\mathbf{v}_{\star}$. Thus, (35) indicates that

$$(\mathbf{B}_j - \mathbf{B}_i) \cdot \mathbf{v}_\star = 0.$$

Since $\mathbf{v}_*$ was arbitrary, $\mathbf{B}_j = \mathbf{B}_i$. Since i and j are also arbitrary, it must be that $\mathbf{B}_i = \mathbf{B}$, for all i , for some vector $\mathbf{B}$ that is i -independent.



Proof (Cont.)

Thus, (35) is equivalent to

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}_\star|^2 - \frac{m_j}{2} |\mathbf{v}'_\star|^2 \right) = 0 \quad (i \leftrightarrow j),$$

for all $\mathbf{v}$, $\mathbf{v}_\star$, and σ . Again, let $\mathbf{v}_\star$ be an arbitrary, non-zero vector. Choose

$$\mathbf{v} = \frac{\mu_{ij} - 2}{\mu_{ij}} \mathbf{v}_\star, \quad \sigma = \frac{\mathbf{v}_\star}{|\mathbf{v}_\star|}.$$

Then, it follows that $\mathbf{v}'_\star = \mathbf{0}$. Thus,

$$(A_j - A_i) |\mathbf{v}_\star|^2 = 0.$$

As above, it must be that $A_i = A$, for all i , where A is i -independent. ■



Moments and Moment Equations



In this section, we develop the macroscopic moment equations from the Boltzmann equations.

Definition

Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, R$ are kinetic distributions with the property that

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, \quad (37)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Assume that m_i , the **mass of the i^{th} particle**, is a positive real number. The quantity

$$n_i(\mathbf{x}, t) := \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) d\mathbf{v} \quad (38)$$

is called the **number density of the i^{th} species**, while

$$\rho_i(\mathbf{x}, t) := m_i n_i(\mathbf{x}, t) \quad (39)$$

is called the **mass density of the i^{th} species**.



Definition

The quantity

$$\mathbf{u}_i(\mathbf{x}, t) := \frac{1}{n_i(\mathbf{x}, t)} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} \, d\mathbf{v} \quad (40)$$

is called the **bulk velocity** the i^{th} species. The **total mass density of the mixture** is defined as

$$\rho(\mathbf{x}, t) := \sum_{i=1}^R \rho_i(\mathbf{x}, t). \quad (41)$$

The quantity

$$\mathbf{q}(\mathbf{x}, t) := \sum_{i=1}^R \rho_i(\mathbf{x}) \mathbf{u}_i(\mathbf{x}, t) \quad (42)$$

is called the **total momentum density of the mixture**. The **bulk mass-averaged velocity** is defined as

$$\mathbf{u}(\mathbf{x}, t) := \frac{\mathbf{q}(\mathbf{x}, t)}{\rho(\mathbf{x}, t)}. \quad (43)$$



Definition

The **kinetic energy density of the i^{th} species** is defined as

$$\omega_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \quad (44)$$

The **total kinetic energy density** is defined as

$$\omega(\mathbf{x}, t) := \sum_{i=1}^R \omega_i(\mathbf{x}, t). \quad (45)$$

The **internal energy density of the i^{th} species** is defined via

$$e_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v} - \mathbf{u}_i(\mathbf{x}, t)|^2 d\mathbf{v}. \quad (46)$$

The **total internal energy density** is defined as

$$e(\mathbf{x}, t) := \sum_{i=1}^R e_i(\mathbf{x}, t). \quad (47)$$



Definition

Finally, the **temperature of the i^{th} species** is defined via

$$\theta_i(\mathbf{x}, t) := \frac{2}{3} \frac{e_i(\mathbf{x}, t)}{n_i(\mathbf{x}, t)}, \quad (48)$$

so that, after some simple manipulation,

$$\theta_i(\mathbf{x}, t) = \frac{m_i}{3n_i(\mathbf{x}, t)} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v} - \mathbf{u}_i(\mathbf{x}, t)|^2 d\mathbf{v}. \quad (49)$$



Remark

Clearly, θ_i , the temperature of the i^{th} species is non-negative, provided f_i is non-negative.

Remark

*The quantities ρ_i , $\mathbf{u}_i$, and θ_i are sometimes referred to, collectively, as the **first three moments of f_i** . Other authors use the collection ρ_i , $\rho_i \mathbf{u}_i$, and ω_i to stand for first three moments of f_i . Observe that the collections $(\rho_i, \mathbf{u}_i, \theta_i)$ and $(\rho_i, \rho_i \mathbf{u}_i, \omega_i)$ are essentially equivalent. One collection of variables can be derived from the other and vice versa.*



Let us define a few more bulk/macroscopic quantities.

Definition

Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, R$ are kinetic distributions with the properties

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (50)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. The **energy flow of the i^{th} species** is defined as

$$\mathbf{r}_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} |\mathbf{v}|^2 d\mathbf{v}. \quad (51)$$

The **i^{th} momentum flow tensor** is defined as

$$\mathbf{M}_i(\mathbf{x}, t) := m_i \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} \otimes \mathbf{v} d\mathbf{v}. \quad (52)$$

The **i^{th} deviation velocity** is defined as

$$\mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) := \mathbf{v} - \mathbf{u}_i(\mathbf{x}, t). \quad (53)$$



Definition

The **pressure tensor of the i^{th} species** is defined as

$$P_i(\mathbf{x}, t) := m_i \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) \otimes \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) d\mathbf{v}. \quad (54)$$

Define the **i^{th} heat flux** via

$$\chi_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{c}_i(\mathbf{x}, \mathbf{v}, t)|^2 d\mathbf{v}. \quad (55)$$

Next, we list some elementary identities that relate our defined bulk/macroscopic quantities.



Proposition

Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, R$, are kinetic distributions with the properties

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (56)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Then, we have the following identities:

$$\omega_i(\mathbf{x}, t) = \frac{\rho_i(\mathbf{x}, t)}{2} |\mathbf{u}_i(\mathbf{x}, t)|^2 + e_i(\mathbf{x}, t); \quad (57)$$

$$\mathbf{M}_i(\mathbf{x}, t) = \rho_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t) \otimes \mathbf{u}_i(\mathbf{x}, t) + \mathbf{P}_i(\mathbf{x}, t); \quad (58)$$

and

$$\mathbf{r}_i(\mathbf{x}, t) = \omega_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t) + \chi_i(\mathbf{x}, t) + \mathbf{P}_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t). \quad (59)$$

Proof.

The proofs are straightforward and are skipped for brevity. ■



Finally, we can state the basic moment equations for the multi-species Boltzmann system.

Proposition

Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, R$, are kinetic distributions that satisfy the multi-species Boltzmann system,

$$\partial_t f_i(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i(\mathbf{x}, \mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t), \quad (60)$$

where $Q_i[f_i, \mathbf{f}]$ is the cumulative binary collision operator

$$Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) := \sum_{j=1}^R Q_{ij}[f_i, f_j](\mathbf{x}, \mathbf{v}, t). \quad (61)$$

Assume that

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (62)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Suppose that the collision kernels, B_{ij} have proper structure.



Then, for all $\mathbf{x} \in \Omega$ and $t \geq 0$, and for each $i = 1, \dots, R$,

$$\partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i) = m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \mathbf{v} d\mathbf{v}, \quad (64)$$

and

$$\begin{aligned} \partial_t \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i \right] + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i) \\ = \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \end{aligned} \quad (65)$$



Consequently, using the collisional invariants,

$$\partial_t \rho + \nabla_x \cdot (\rho \mathbf{u}) = 0, \quad (66)$$

$$\partial_t(\rho \mathbf{u}) + \nabla_x \cdot \sum_{i=1}^R (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i) = 0, \quad (67)$$

and

$$\partial_t \sum_{i=1}^R \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \sum_{i=1}^R \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i + \boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i \right] = 0. \quad (68)$$

Proof.

To prove the first three identities, multiply (60) by, respectively, m_i , $m_i \mathbf{v}$, and $\frac{m_i}{2} |\mathbf{v}|^2$ and integrate over velocity space. The last three are obtained by summing the first three on i and using the usual collision invariants (16) – (18).

Proof (Cont.)



Here we will prove only identities (64) and (65) in detail. First, multiply (60) by $m_i \mathbf{v}$ and integrate over velocity space to obtain

$$\begin{aligned} m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \mathbf{v} \, d\mathbf{v} &= \partial_t (\rho_i \mathbf{u}_i) + m_i \int_{\mathbb{R}^3} \mathbf{v} \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i \, d\mathbf{v} \\ &= \partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot \mathbf{M}_i \\ &= \partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i), \end{aligned}$$

where we have used identity (58). Thus, we have identity (64).

Next, multiply (60) by $\frac{m_i}{2} |\mathbf{v}|^2$ and integrate over velocity space to obtain

$$\begin{aligned} \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 \, d\mathbf{v} &= \partial_t \omega_i + \nabla_{\mathbf{x}} \cdot \mathbf{r}_i \\ &= \partial_t \omega_i + \nabla_{\mathbf{x}} \cdot (\omega_i \mathbf{u}_i + \boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i) \\ &= \partial_t \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i \right] \\ &\quad + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i), \end{aligned}$$

using (59) and (57), which gives the identity (65). ■



The Maxwellian and Maximal Entropy States



In this section, we will do our calculations in a physical space of generic dimension $d \in \mathbb{N}$, to maintain some generality.

Definition

Suppose that $\Omega \subset \mathbb{R}^d$ is an open, bounded set and $f : \Omega \times \mathbb{R}^d \times [0, \infty) \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass $m > 0$, with the property

$$\int_{\mathbb{R}^d} f(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, \quad (69)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. The **Maxwellian distribution** of f is defined as

$$M_f(\mathbf{x}, \mathbf{v}, t) := M[f](\mathbf{x}, \mathbf{v}, t) := n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2\theta} \right), \quad (70)$$

using the first three integral moments of f ,

$$n(\mathbf{x}, t) := \int_{\mathbb{R}^d} f d\mathbf{v}, \quad \mathbf{u}(\mathbf{x}, t) := \frac{1}{n} \int_{\mathbb{R}^d} f \mathbf{v} d\mathbf{v}, \quad \theta(\mathbf{x}, t) := \frac{m}{d n} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 d\mathbf{v}. \quad (71)$$



For convenience, we define the bracket notation, $\langle \cdot \rangle: L^1(\mathbb{R}^d; \mathbb{R}^p) \rightarrow \mathbb{R}^p$, by

$$\langle \mathbf{g} \rangle = \int_{\mathbb{R}^d} \mathbf{g}(\mathbf{v}) \, d\mathbf{v}.$$

Therefore, we have the usual definitions

$$\begin{bmatrix} \rho(\mathbf{x}, t) \\ \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \\ \omega(\mathbf{x}, t) \end{bmatrix} = \left\langle f(\mathbf{x}, \mathbf{v}, t) \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2} |\mathbf{v}|^2 \end{bmatrix} \right\rangle,$$

where $m > 0$ is the mass of the particles with kinetic distribution f .

Bracket Notation



For convenience, let us define

$$\boldsymbol{\rho}(\mathbf{x}, t) := \left[\rho, \rho \mathbf{u}^\top, \omega \right]^\top : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}^{d+2}; \quad (72)$$

$$\mathbf{m}(\mathbf{v}) := \left[m, m \mathbf{v}^\top, \frac{m}{2} |\mathbf{v}|^2 \right]^\top : \mathbb{R}^d \rightarrow \mathbb{R}^{d+2}; \quad (73)$$

and

$$\boldsymbol{\alpha}(\mathbf{x}, t) := \frac{1}{m} \left[\ln \left(\frac{n}{(2\pi\theta/m)^{d/2}} \right) - \frac{m|\mathbf{u}|^2}{2\theta}, \frac{m\mathbf{u}^\top}{\theta}, -\frac{m}{\theta} \right]^\top : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}^{d+2}. \quad (74)$$



Then, succinctly,

$$\rho = \langle f \mathbf{m} \rangle, \quad \text{and} \quad M_f = \exp(\boldsymbol{\alpha} \cdot \mathbf{m}(\mathbf{v})). \quad (75)$$

Indeed,

$$\begin{aligned} \exp(\boldsymbol{\alpha} \cdot \mathbf{m}) &= \exp\left(\ln\left(\frac{n}{(2\pi\theta/m)^{d/2}}\right) - \frac{m|\mathbf{u}|^2}{2\theta}\right) \exp\left(\frac{m\mathbf{u} \cdot \mathbf{v}}{\theta}\right) \exp\left(-\frac{m}{2\theta}|\mathbf{v}|^2\right) \\ &= n \left(\frac{m}{2\pi\theta}\right)^{d/2} \exp\left(-\frac{m|\mathbf{u}|^2}{2\theta}\right) \exp\left(\frac{m\mathbf{u} \cdot \mathbf{v}}{\theta}\right) \exp\left(-\frac{m}{2\theta}|\mathbf{v}|^2\right) \\ &= n \left(\frac{m}{2\pi\theta}\right)^{d/2} \exp\left(-\frac{m}{2\theta}|\mathbf{u} - \mathbf{v}|^2\right) \\ &= M_f. \end{aligned}$$



The next result indicates that the first three moments of f and its associated Maxwellian are exactly the same.

Lemma

Suppose that $f : \mathbb{R}^d \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass $m > 0$, with the property

$$\int_{\mathbb{R}^d} f(\mathbf{v}) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2. \quad (76)$$

The following equality holds:

$$\langle (M_f - f) \mathbf{m} \rangle = \left\langle \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2} |\mathbf{v}|^2 \end{bmatrix} (M_f - f) \right\rangle = \mathbf{0}. \quad (77)$$



(1) First we show that

By definition,

$$\begin{aligned} \int_{\mathbb{R}^d} M_f \, d\mathbf{v} &= \int_{\mathbb{R}^d} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \exp\left(-\frac{m|\mathbf{v}-\mathbf{u}|^2}{2\theta}\right) d\mathbf{v} \\ &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \exp\left(-\left|\frac{\mathbf{v}-\mathbf{u}}{(2\theta/m)^{\frac{1}{2}}}\right|^2\right) d\mathbf{v}. \end{aligned} \quad (79)$$



Proof (Cont.)

Consider the substitution

$$\mathbf{s} = \frac{\mathbf{v} - \mathbf{u}}{(2\theta/m)^{\frac{1}{2}}} \implies d\mathbf{s} = \left(\frac{1}{(2\theta/m)^{\frac{1}{2}}} \right)^d d\mathbf{v} \implies d\mathbf{v} = (2\theta/m)^{\frac{d}{2}} d\mathbf{s}. \quad (80)$$

The integral becomes

$$\begin{aligned} \int_{\mathbb{R}^d} M_f d\mathbf{v} &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \exp(-|\mathbf{s}|^2) (2\theta/m)^{\frac{d}{2}} d\mathbf{s} \\ &= \frac{n(2\theta/m)^{\frac{d}{2}}}{\pi^{\frac{d}{2}} (2\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s} \\ &= \frac{n}{\pi^{\frac{d}{2}}} \cdot \pi^{\frac{d}{2}} \\ &= \int_{\mathbb{R}^d} f d\mathbf{v}, \end{aligned} \quad (81)$$

as desired. Note that we have used the definition $n = \int_{\mathbb{R}^d} f d\mathbf{v}$, and the fact that in d dimensions, the integral $\int_{\mathbb{R}^d} e^{-|\mathbf{x}|^2} d\mathbf{x} = \pi^{\frac{d}{2}}$.



Proof (Cont.)

(2) Next we show that

$$\int_{\mathbb{R}^d} \mathbf{v} M_f \, d\mathbf{v} = \int_{\mathbb{R}^d} \mathbf{v} f \, d\mathbf{v}. \quad (82)$$

Following a similar line of work, we have

$$\begin{aligned} \int_{\mathbb{R}^d} \mathbf{v} M_f \, d\mathbf{v} &= \int_{\mathbb{R}^d} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \mathbf{v} \exp\left(-\frac{|\mathbf{v} - \mathbf{u}|^2}{2\theta/m}\right) d\mathbf{v} \\ &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \mathbf{v} \exp\left(-\left|\frac{\mathbf{v} - \mathbf{u}}{(2\theta/m)^{\frac{1}{2}}}\right|^2\right) d\mathbf{v}. \end{aligned} \quad (83)$$



Proof (Cont.)

Using the substitution in (80), the integral becomes

$$\begin{aligned}
 \int_{\mathbb{R}^d} \mathbf{v} M_f \, d\mathbf{v} &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \left((2\theta/m)^{\frac{1}{2}} \mathbf{s} + \mathbf{u} \right) \exp(-|\mathbf{s}|^2) \cdot (2\theta/m)^{\frac{d}{2}} \, d\mathbf{s} \\
 &= \frac{n(2\theta/m)^{\frac{d}{2}}}{\pi^{\frac{d}{2}} (2\theta/m)^{\frac{d}{2}}} \left[(2\theta/m)^{\frac{1}{2}} \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} \, d\mathbf{s} + \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} \, d\mathbf{s} \right] \\
 &= \frac{n}{\pi^{\frac{d}{2}}} \left[(2\theta/m)^{\frac{1}{2}} \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} \, d\mathbf{s} + \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} \, d\mathbf{s} \right]. \quad (84)
 \end{aligned}$$

Let us deal with the first term in (84). One can easily show that

$$\int_{-\infty}^{\infty} x e^{-x^2} \, dx = 0 \implies \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} \, d\mathbf{s} = \mathbf{0}.$$

Therefore, we have

$$\int_{\mathbb{R}^d} \mathbf{v} M_f \, d\mathbf{v} = \frac{n}{\pi^{\frac{d}{2}}} \cdot \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} \, d\mathbf{s} = \frac{n\mathbf{u}}{\pi^{\frac{d}{2}}} \cdot \pi^{\frac{d}{2}} = n\mathbf{u} = \int_{\mathbb{R}^d} \mathbf{v} f \, d\mathbf{v}.$$



Proof (Cont.)

(3) Finally, we show that

$$\int \frac{|\mathbf{v}|^2}{2} M_f \, d\mathbf{v} = \int \frac{|\mathbf{v}|^2}{2} f \, d\mathbf{v}. \quad (85)$$

First, observe that

$$\int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} = \frac{d}{2} \pi^{\frac{d}{2}}. \quad (86)$$

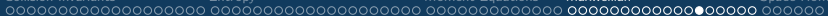


Proof (Cont.)

Now, using the substitution in (80), and the same integration techniques on the LHS of (85),

$$\begin{aligned}
 \int \frac{|\mathbf{v}|^2}{2} M_f \, d\mathbf{v} &= \frac{1}{2} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int |\mathbf{v}|^2 e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} \, d\mathbf{v} \\
 &= \frac{1}{2} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int |\mathbf{u} + (\mathbf{v} - \mathbf{u})|^2 e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} \, d\mathbf{v} \\
 &= \frac{n}{2(2\pi\theta/m)^{\frac{d}{2}}} \int \left(-|\mathbf{u}|^2 + 2\mathbf{u}^\top \mathbf{v} + |\mathbf{v} - \mathbf{u}|^2 \right) e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} \, d\mathbf{v} \\
 &= \frac{-n|\mathbf{u}|^2}{2\pi^{\frac{d}{2}}} \pi^{\frac{d}{2}} + \frac{n|\mathbf{u}|^2}{\pi^{\frac{d}{2}}} \pi^{\frac{d}{2}} + 0 + \frac{2n\theta/m}{2\pi^{\frac{d}{2}}} \frac{d}{2} \pi^{\frac{d}{2}} \\
 &= \frac{1}{2} n|\mathbf{u}|^2 + \frac{d}{2} n\theta/m \\
 &= \int_{\mathbb{R}^d} \frac{|\mathbf{v}|^2}{2} f \, d\mathbf{v},
 \end{aligned} \tag{87}$$

where the final line follows from identity (57) (with d and 3 interchanged). The proof is complete. ■



We need the following simple result.

Lemma

Suppose that $f : \mathbb{R}^d \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass $m > 0$, with the property

$$\int_{\mathbb{R}^d} f(\mathbf{v}) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2. \quad (88)$$

The function $\ln(M_f)$ is a Maxwellian invariant, namely,

$$\int_{\mathbb{R}^d} \ln(M_f(\mathbf{v})) (M_f(\mathbf{v}) - f(\mathbf{v})) \, d\mathbf{v} = \langle \ln(M_f)(M_f - f) \rangle = 0.$$



Proof.

From the last result, we already know that, for any $C_0, C_2 \in \mathbb{R}$, and any $\mathbf{C}_1 \in \mathbb{R}^d$,

$$\langle (C_0 + \mathbf{C}_1 \cdot \mathbf{v} + C_2 |\mathbf{v}|^2)(M_f - f) \rangle = 0.$$

Clearly,

$$\ln(M_f) = \ln(\exp(\boldsymbol{\alpha} \cdot \mathbf{m})) = \boldsymbol{\alpha} \cdot \mathbf{m},$$

is a linear combination of 1, $\mathbf{v}$, and $|\mathbf{v}|^2$. It follows that $\ln(M_f)$ is also a Maxwellian invariant, that is, $\langle \ln(M_f)(M_f - f) \rangle = 0$. ■



The Maxwellian is the entropy maximizing state. Consider the following result.

Theorem

Suppose that $f_\star : \mathbb{R}^d \rightarrow [0, \infty)$ is a given kinetic distribution. Let $\boldsymbol{\rho}_\star$ be the vector comprised of the first three moments of $f_\star$, that is,

$$\boldsymbol{\rho}_\star = \begin{bmatrix} \rho_\star \\ \rho_\star \mathbf{u}_\star \\ \omega_\star \end{bmatrix} = \left\langle f_\star(\mathbf{v}) \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2}|\mathbf{v}|^2 \end{bmatrix} \right\rangle.$$

The temperature, $\theta_\star$, is defined via

$$\omega_\star = \frac{\rho_\star}{2} |\mathbf{u}_\star|^2 + e_\star = \frac{\rho_\star}{2} |\mathbf{u}_\star|^2 + \frac{d}{2} n_\star \theta_\star.$$

The Maxwellian, as usual, is defined using the macroscopic densities as

$$M_\star(\mathbf{x}, \mathbf{v}) := n_\star \left(\frac{m}{2\pi\theta_\star} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}_\star|^2}{2\theta_\star} \right).$$



Theorem (Cont.)

Consider the admissible class of distribution functions, defined as

$$\mathcal{A}_\star := \left\{ g : \mathbb{R}^d \rightarrow \mathbb{R} \mid g \geq 0 \quad \text{and} \quad \langle \mathbf{m}g \rangle = \boldsymbol{\rho}_\star \right\},$$

where, recall,

$$\mathbf{m}(\mathbf{v}) := \left[m, m\mathbf{v}^\top, \frac{m}{2}|\mathbf{v}|^2 \right]^\top : \mathbb{R}^d \rightarrow \mathbb{R}^{d+2}.$$

Then, $M_\star, f_\star \in \mathcal{A}_\star$. In other words, $\mathcal{A}_\star$ is non-empty. For every $g \in \mathcal{A}_\star$,

$$s[M_\star] \geq s[g],$$

where $s[g]$ is the entropy of g , defined as

$$s[g] := - \int_{\mathbb{R}^d} g(\mathbf{v}) \ln(g(\mathbf{v})) \, d\mathbf{v}.$$

In other words, the Maxwellian is the maximum entropy distribution out of the set of admissible distributions that gives rise to the macroscopic density $\boldsymbol{\rho}_\star$.



Proof.

Let us first show that $M_\star \in \mathcal{A}_\star$. Clearly $M_\star$ is non-negative. The fact that $\langle \mathbf{m} M_\star \rangle = \boldsymbol{\rho}_\star$ follows from a previous simple calculation. Thus $M_\star \in \mathcal{A}_\star$.

Next, we prove the inequality. Let $g \in \mathcal{A}_\star$ be arbitrary. We may write $M_\star = M_g$, since g gives rise to the same macroscopic fields as $f_\star$. Define $\hat{h}(s) := s \ln(s)$. This function $\hat{h}$ is convex on $(0, \infty)$, as is easy to show. Its first two derivatives are

$$\hat{h}'(s) = \ln(s) + 1, \quad \hat{h}''(s) = \frac{1}{s} > 0, \quad \forall s > 0.$$

Thus,

$$\hat{h}(s) \geq \hat{h}(t) + \hat{h}'(t)(s - t), \quad \forall s, t \in (0, \infty).$$

It follows that, point-wise in $\mathbf{v}$,

$$\begin{aligned} \hat{h}(g) &\geq \hat{h}(M_g) + \hat{h}'(M_g)(g - M_g) \\ &= \hat{h}(M_g) + \ln(M_g)(g - M_g) + g - M_g. \end{aligned}$$



Proof (Cont.)

By a previous lemma,

$$\begin{aligned}\int_{\mathbb{R}^d} \hat{h}(g) \, d\mathbf{v} &\geq \int_{\mathbb{R}^d} \hat{h}(M_g) \, d\mathbf{v} + \int_{\mathbb{R}^d} \ln(M_g)(g - M_g) \, d\mathbf{v} + \int_{\mathbb{R}^d} (g - M_g) \, d\mathbf{v} \\ &= \int_{\mathbb{R}^d} \hat{h}(M_g) \, d\mathbf{v}.\end{aligned}$$

This shows that, reversing signs,

$$s[g] \leq s[M_g] = s[M_*],$$

and the result is proven. ■



Space-Homogeneous Model



Definition

We say that the kinetic distributions $f_i : \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$, $i = 1, \dots, R$, satisfy the **space-homogeneous multi-species Boltzmann equation** iff, for all $\mathbf{v} \in \mathbb{R}^3$ and $t \in (0, T]$,

$$\partial_t f_i(\mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{v}, t), \quad (89)$$

where $Q_i[f_i, \mathbf{f}]$, the **cumulative binary collision operator**, is defined as

$$Q_i[f_i, \mathbf{f}](\mathbf{v}, t) := \sum_{j=1}^R Q_{i,j}[f_i, f_j](\mathbf{v}, t). \quad (90)$$

For the space-homogeneous model, the approach of the system to the equilibrium Maxwellian states is clear.



Proposition (H-Theorem)

Suppose that $f_1, \dots, f_R : \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the space-homogeneous multi-species Boltzmann equation. Assume that the collision kernels $B_{i,j}$ have proper structure. Then,

$$\partial_t s(t) = \sigma_s(t) \geq 0, \quad (91)$$

where s is the **entropy**, defined as

$$s(t) := \sum_{i=1}^R \int_{\mathbb{R}^3} f_i(\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) \, d\mathbf{v},$$

and σ_s is the **entropy production rate**, defined via

$$\sigma_s(t) := - \sum_{i=1}^R \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) \, d\mathbf{v}.$$

Proof.

The result follows from a previous calculation. ■



The following also follows from a previously-established result.

Proposition

Suppose that $f_i : \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, R$, are kinetic distributions with the properties

$$\int_{\mathbb{R}^3} f_i(\mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (92)$$

for every $t \geq 0$. Assume that the kinetic distributions satisfy the space-homogeneous multi-species Boltzmann system. Then, for all $t \geq 0$, and for each $i = 1, \dots, R$,

$$d_t \rho_i(t) = 0, \quad (93)$$

$$\rho_i d_t \mathbf{u}_i(t) = m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) \mathbf{v} d\mathbf{v}, \quad (94)$$

and

$$d_t \left(\frac{\rho_i}{2} |\mathbf{u}_i(t)|^2 + e_i(t) \right) = \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \quad (95)$$



Proposition (Cont.)

Consequently, using the collisional invariants,

$$d_t \rho(t) = 0, \quad (96)$$

$$\rho \, \mathrm{d}_t \mathbf{u}(t) = 0, \quad (97)$$

and

$$d_t \omega(t) = d_t \sum_{i=1}^R \left(\frac{\rho_i}{2} |\mathbf{u}_i(t)|^2 + e_i(t) \right) = 0. \quad (98)$$



Remark

It can be proven that solutions to the space-homogeneous multi-species Boltzmann equation increase the entropy,

$$s(t) := - \sum_{i=1}^R \int_{\mathbb{R}^3} f_i(\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) \, d\mathbf{v},$$

until the system reaches an equilibrium state (at $t = \infty$) characterized by

$$0 = [\sigma_s]_{\infty} := - \sum_{i=1}^R \int_{\mathbb{R}^3} Q_i[f_{i,\infty}, \mathbf{f}_{\infty}](\mathbf{v}) \ln(f_{i,\infty}(\mathbf{v})) \, d\mathbf{v}.$$

The kinetic distributions evolve in such a way that p_i is time invariant, and the total momentum and energy are conserved. At equilibrium, the kinetic distributions will be Maxwellians, as guaranteed by a previous proposition. The particle temperatures will equilibrate to the same temperatures, that is,

$$\lim_{t \rightarrow \infty} \theta_i(t) = \theta_{\infty} > 0, \quad i = 1, \dots, R.$$